Network’s Lab.

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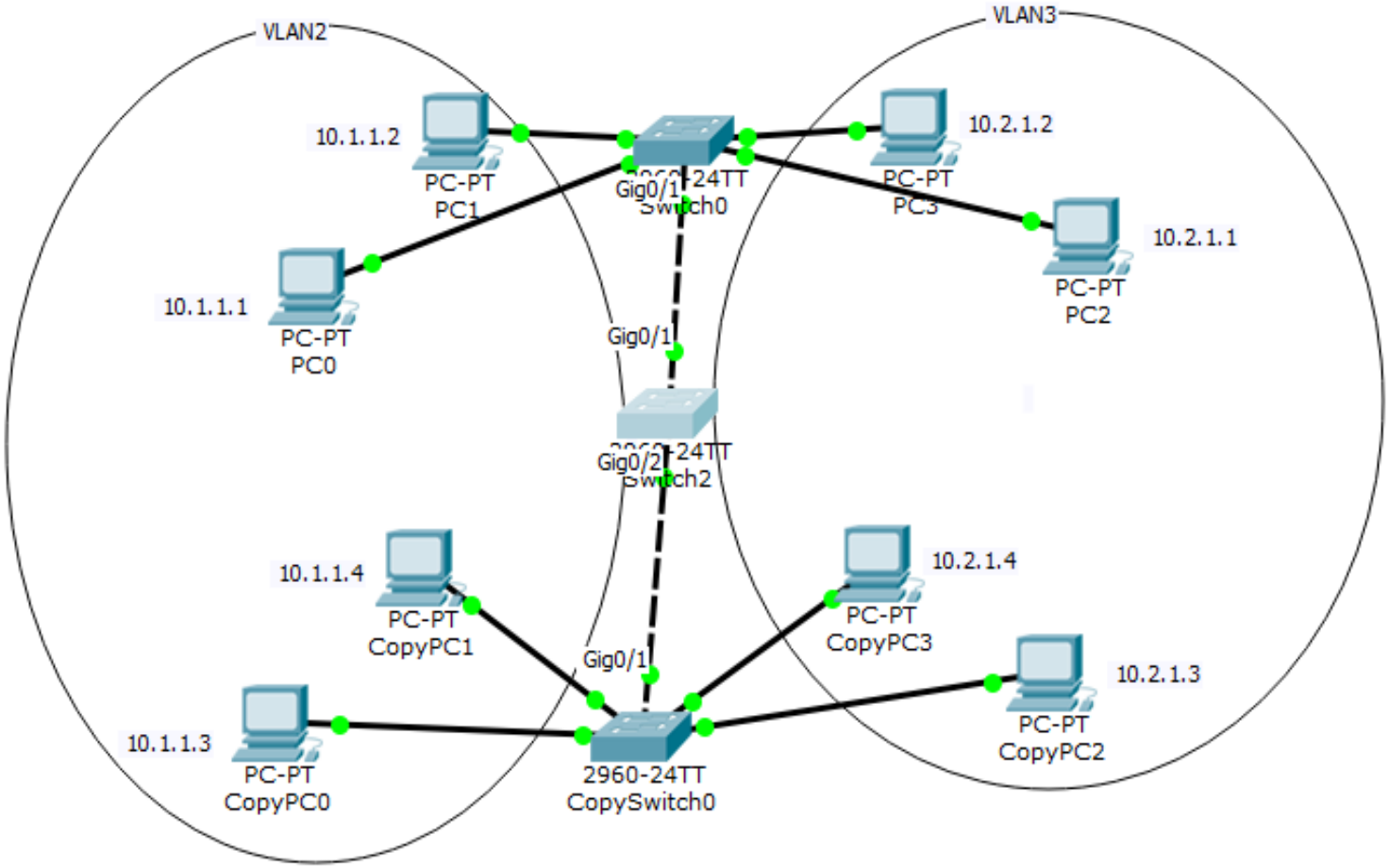
Session: 6

S6_1.Vlan in 3 switches

- Based on S5_2.
  - Add 3rd switch

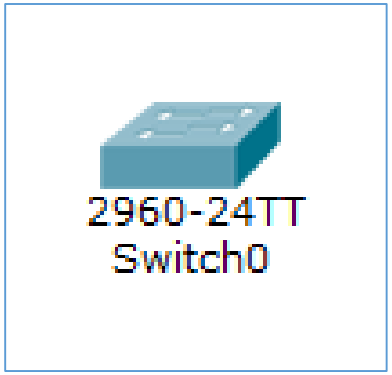


- Try to ping from above to below at the same Vlan

S6_1.Vlan in 3 switches

- At the new switch
  - **#show interfaces trunk**
    - See, two GE interfaces are in ***trunking mode*** (they are by default in dynamic mode)
    - You can force them to stay in trunk mode *(i.e. switchport mode trunk)*
    - Also see, ***Vlans allowed and active in management domain***
    - Only Vlan1 is active at the switch
  - Add two vlan 2 and 3 to make them active
    - (config)#vlan 2
    - (config-vlan)#exit
    - **Add Vlan 3 as well**
  - **#show interfaces trunk**
- Try to ping from above to below at the same Vlan

S6_2. Interface Range

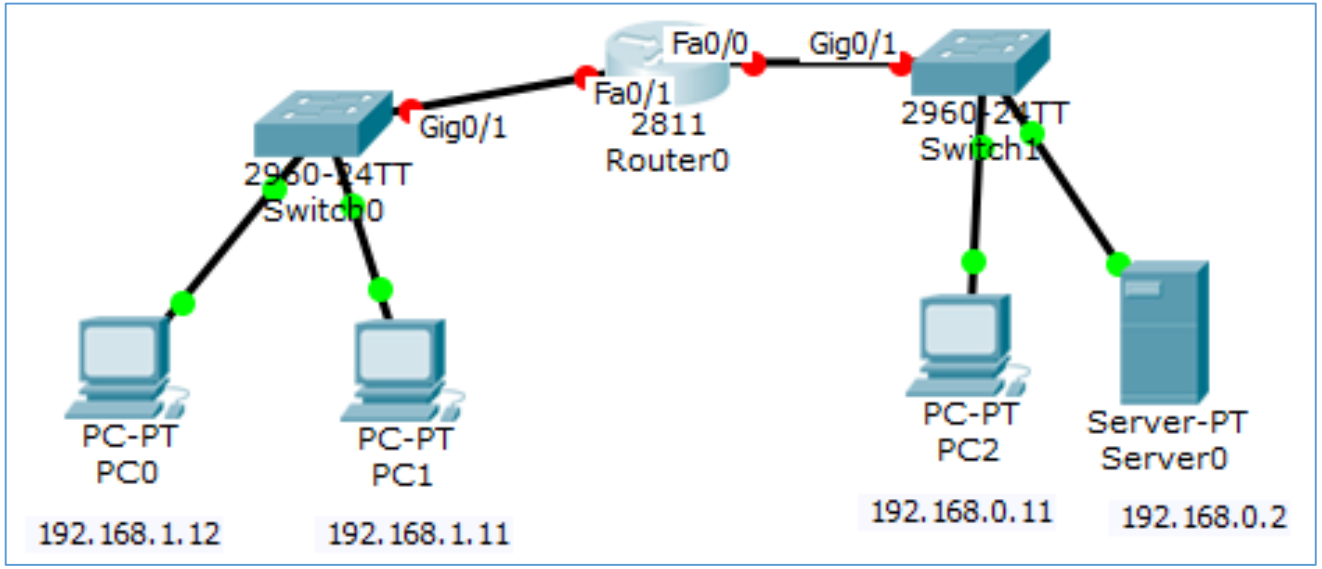


- Add a single switch:
- #show vlan
- We want to assign
  - FE 1 to 10 to VLAN 2,
  - FE 11 to 15 to VLAN 3,
  - FE 16 to 24 to VLAN 4
- (config)#vlan 2
- (config-vlan)#exit
- **Add Vlan 3 and 4 as well**
- (config)#interface range fastEthernet 0/1-10
- (config-if-range)#switchport mode access
- (config-if-range)#switchport access vlan 2
- (config-if-range)#exit
- **Do that for *fastEthernet 0/11-15* and *fastEthernet 0/16-24***
- #show vlan

S6_2. Interface Range

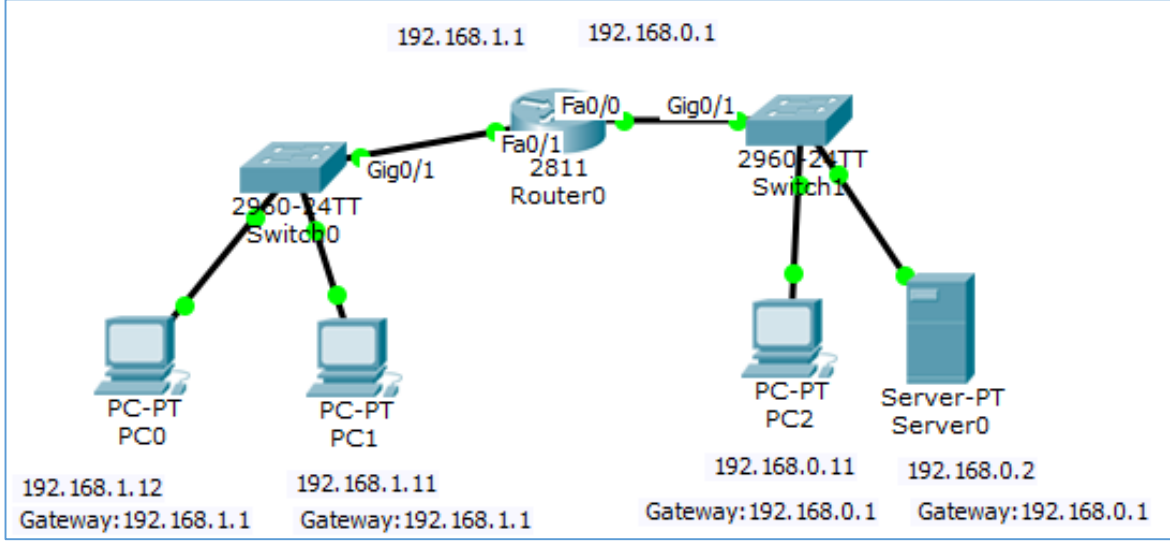
- Add computers and set their IP from 192.168.0.*/24
- Check if computers connecting to a same VLAN can ping each other.

S6_3. One router



- Try ping inside each network, try ping between networks
- Note that Router's ports are in shutdown mode by default
- Router#show interfaces
- Notice that port numbers start from 0
  - FastEthernet0/0 , FastEthernet0/1

S6_3. One router



- We need to assign **IP** to each FE port in router and turn it on
  - Router#conf t
  - Router(config)#interface fastEthernet 0/0
  - Router(config-if)#ip address 192.168.0.1 255.255.255.0
  - Router(config-if)#no shutdown
  - Router(config-if)#exit
  - Router(config)#interface fastEthernet 0/1
  - Router(config-if)#ip address 192.168.1.1 255.255.255.0
  - Router(config-if)#no shutdown
  - Router(config-if)#exit
- Ping From router to PCs and From PCs to Gateway address
- Ping Between Computers in two different networks
  - It works because Two networks are directly connected to the router's ports
- Router#show ip interface brief
- Router#show ip route